

BORA UNER

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EDUCATION

Michigan State University <i>B.S. Computer Science & Economics; GPA: 3.93/4.0</i> - Honors College; Dean's List (all semesters) - Certification: Algolia Certified Developer (ACD) — In Progress	East Lansing, MI Aug 2023 – May 2027
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EXPERIENCE

Software Engineering Intern <i>Corewell Health</i> - Strengthened corewellhealth.org's disaster recovery by building region-replicated API Gateway stacks and Route 53 health-based traffic routing for automated failover, working under the Dev Team Lead. - Built a dynamic "Related Pages" feature for location pages that maps each page's taxonomy terms to a service-line dictionary, surfacing relevant service links to improve content discoverability. - Replaced legacy breadcrumb navigation with cleaner regional location badges using TypeScript and SCSS, improving UX consistency and consumer wayfinding across site pages. - Contributed to Agile sprints through standups, planning, and retrospectives, collaborating closely with engineers, designers, and interns to deliver user-focused solutions.	May 2026 – Present Grand Rapids, MI
Resident Assistant <i>MSU Student Life & Engagement</i> - Fostered an inclusive living environment for 48 residents through academic programming and conflict resolution initiatives. - Mentored peers and connected them with campus organizations to enhance engagement.	Aug 2025 – Present East Lansing, MI
Undergraduate Learning Assistant, Physics <i>MSU College of Natural Sciences</i> - Led student groups applying physics to real-world problems and delivered tailored weekly feedback. - Strengthened collaboration by refining instruction methods with fellow Learning Assistants.	Jan 2024 – Jan 2025 East Lansing, MI
Engineering Logistics Team <i>MSU College of Engineering</i> - Supported first-year engineering students in academic planning and transition to college life. - Designed promotional materials using Canva and Photoshop for outreach events.	Apr 2024 – Jan 2025 East Lansing, MI

PROJECTS

Object-Oriented Programming Projects - Developed two advanced C++ applications: a Mastermind-style puzzle game and an animated machine simulation. - Applied OOP principles and implemented design patterns including Factory, Visitor, Adapter, Observer, Composite, and Iterator. - Built interactive GUIs and animations using wxWidgets with event handling, custom rendering, and keyframe animation. - Led agile sprints for a 7-member team, facilitating collaboration with UML modeling and coding standards.	January 2025 - August 2025
High-Performance Bit Matrix Rotation - Engineered a block-based rotation algorithm for large bit matrices, optimizing cache usage with loop unrolling and prefetching. - Accelerated block transformations by vectorizing with AVX2 intrinsics, enabling bit manipulations and reducing memory overhead. - Improved runtime efficiency and scalability across both even and odd matrix sizes.	January 2025 - April 2025
Parallel Ray Tracing & Physics Simulation - Built a parallelized ray tracer with efficient ray-sphere intersection and Lambertian shading using Cilk parallel loops. - Optimized rendering via projection-based bounding volumes, precomputed lighting, and a custom hybrid quicksort. - Implemented an N-body physics simulation with collision handling and acceleration updates, solving race conditions and scaling efficiently across multicore systems.	January 2025 - August 2025
Social Media Backend Project - Engineered a scalable backend in Python/Flask with secure authentication, follower graph management, and engagement features, designed for production-level deployment. - Architected and optimized a SQLite schema with indexing and normalization to boost performance. - Transitioned from a monolithic design to a Dockerized microservices architecture for scalability and maintainability.	August 2024 - January 2025

SKILLS

Programming Languages: Python, TypeScript, JavaScript, C, C++, SQL, HTML, CSS, SCSS Frontend: React, Gatsby, GraphQL, Contentstack (Headless CMS) Backend & Cloud: AWS (CDK, Lambda, API Gateway, S3, CloudFront, Route 53, Kendra), Algolia, REST APIs, Flask, Microservices Systems & Tools: Docker, Git, Jira, Linux, CI/CD	Software Engineering: Object-Oriented Programming, Agile/Scrum, Design Patterns, UML Modeling, Infrastructure as Code, System Design, Unit Testing Core Skills: Parallel Programming, Performance Optimization, Data Structures & Algorithms, Database Schema Design Leadership & Collaboration: Team Leadership, Mentorship, Teaching, Research, Interdisciplinary Communication
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